

**CO10504: Computer Programming**  
**Lab Assignment #4**

**Functions and Recursion**

1. Write a program in C to find the sum of the series  $1!/1+2!/2+3!/3+4!/4+5!/5$  using the function.
2. Write a program in C to convert a decimal number to binary number using the function.
3. Write a program in C to print the first 50 natural numbers using recursion.
4. Write a program in C to find GCD of two numbers using recursion.
5. Write a program in C to calculate the power of any number using recursion.
6. Write a program in C to find the sum of digits of a number using recursion.
7. Write a program in C to find the Hailstone Sequence of a given number upto 1.  
Hailstone Sequence: Start with any positive integer (an initial seed) and obtain a sequence of numbers by following these rules.
  - If the current number is even, divide it by two; else if it is odd, multiply it by three and add one.
  - Repeat.

For example: if the user inputs 13, the output should be:

The hailstone sequence starting at 13 is :

13 40 20 10 5 16 8 4 2 1

The length of the sequence is 10.